Lee Yan Le Ryan

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EDUCATION

National University of Singapore **Aug 2022 – Jun 2026**
Bachelor of Science (Hons)

- Major in Data Science and Analytics
- Minor in Computer Science
- CGPA: 4.31 / 5.0

TECHNICAL SKILLS

- Programming Languages: Python, Java, C++, C#, R, SQL
- Software: Word, PowerPoint, Excel, VSCode, Git, Azure
- Graphics: Photo Editing, Video Editing

EXPERIENCE

AI Engineer Intern, Synapxe, Singapore **Jun 2025 – Jan 2026**

- Built a Python based UDAD tool that classifies authorised and unauthorised access using either rule-based logic or NLP embedding frameworks chosen by the client, achieving approx. 96.8% accuracy on 2643 records in 8 seconds
- Deployed LightRAG on Azure App Service by resolving linux wheelhouse dependency issues and deployment issues, and obtained more accurate and in-depth answers in a shorter response time compared to traditional RAG systems
- Optimised the AI2D feature building pipeline by integrating Polars, cutting loading time from around 282 seconds to 22 seconds on a desktop and from around 443 seconds to 78 seconds on a laptop, achieving approx. 90% speed ups on both machines
- Explored DLP use cases by evaluating local LLMs such as Ollama models for PII detection on strings potentially containing sensitive data, and improved extraction reliability through prompt engineering techniques like few-shot prompting

Teaching Assistant, NUS, Singapore **Aug 2024 – Nov 2024**

- Mentored 25 undergraduates in CS1010E, a course in NUS on computational thinking and problem-solving using Python
- Received 9 nominations for teaching excellence and rated 4.5/5 for overall teaching, slightly better than computing departments
- Designed custom slides, using animations for visual clarity to bridge learning gaps

PROJECTS

Detection of COVID-19 using Chest X-Ray Scans **Aug 2024 – Dec 2024**

- Conducted binary classification on 535 greyscale X-ray images
- Utilised three CNN models (ConvNet, ResNet18, DenseNet121) from PyTorch library
- Applied GradCAM and GradCAM++ to visualize decision-making regions
- Succeeded in detecting COVID-19 with 77.8% accuracy

Breast Cancer Analysis **Feb 2024 – Apr 2024**

- Conducted binary classification on 569 labelled samples
- Detected mislabels using feature engineering, clustering and logistic regression
- Utilised four Machine Learning models (LR, kNN, RF, SVM) from sklearn library
- Succeeded in detecting benign and malignant tumours with 96% accuracy